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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - August 2025

21EC502 - MICROPROCESSOR AND MICROCONTROLLER

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the architecture of 8051 microcontroller with a neat diagram.	12	L*2	1	1
b)	Assume that register A has packed BCD number 29, write an assembly level program to convert packed BCD to two ASCII numbers and place them in R2 and R6.	8	L3	1	1
2. a)	With a neat diagram explain 128 bytes RAM, organization of 8051 microcontroller with address in detail.	12	L2	1	1
b)	Explain the working of stack using PUSH and POP instructions.	8	L2	1	1
3. a)	Define addressing mode. Explain different types of addressing modes with two examples for each.	12	L2	1	1
b)	For 8051 system of 11.0592 MHz, find how long it takes to execute each instruction. (i) MOV R3, #55 (ii) DEC R3 (iii) DJNZ R2 target (iv) LJMP (v) SJMP (vi) NOP (vii) MUL AB.	8	L3	1	1
Unit – II					
4. a)	Explain TMOD and SCON register of 8051 microcontroller. Assume that XTAL = 11.0592 MHz, write a program to generate a square wave of 2 kHz frequency on pin P1.5. Use 16bit mode.	12	L3	2	1
b)	Explain TCON Register of 8051 microcontrollers. Write an 8051 C program to toggle all the bits of port P1 continuously with some delay in between. Use Timer 0, 16-bit mode to generate the delay.	8	L3	2	1
5. a)	Define interrupt. Explain different types of interrupts in 8051 microcontroller.	8	L2	2	1
b)	Explain (i) SPI Protocol (ii) I2C Protocol.	12	L2	2	1
6. a)	Explain (i) SBUF (ii) TI (iii) RI (iv) Baud Rate. Write a program for the 8051 to transfer "YES" serially at 9600 baud, 8-bit data, 1 stop bit, do this continuously.	12	L3	3	1
b)	Write an 8051 C program to toggle all the bits of P0, P1 and P2 continuously with a 250 ms delay. Use the sfr keyword to declare the port addresses.	8	L3	3	1
Unit – III					
7. a)	Explain the ARM architecture with a neat diagram.	10	L2	4	1
b)	Explain the structure of 5 stage pipeline organization of ARM processor in detail.	10	L2	5	1
8. a)	Explain different Shift and Rotate instructions in ARM with an example for each.	10	L2	4	1
b)	Explain different features of ARM.	10	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
